

TEDDY S. RAUCHWAY

Ithaca, NY | (720) 275-9106 | tsr53@cornell.edu | trauchway.github.io

EDUCATION

Cornell University, Ithaca, NY

Expected May 2028

College of Engineering

Bachelor of Science in Mechanical Engineering

Minor in Aerospace Engineering

GPA: 3.53/4.00

NCAA Division 1 Track & Field Team

Northfield High School, Denver, CO

May 2024

International Baccalaureate Diploma

ENGINEERING EXPERIENCE

Mechanical Engineering Intern

Jun 2026 – Aug 2026

ICR, Inc | Mason, OH

- Designed a fully manufacturable multi-layer protective enclosure for classified defense equipment, addressing environmental, security, and integration requirements across deployment sites
- Developed complete CAD models and detailed engineering drawings in SolidWorks to communicate structural, sealing, and fastening solutions to the manufacturing team
- Performed CFD and FEA simulations using premium analysis tools to optimize airflow, thermal behavior, and structural integrity under operational conditions
- Led the entire design effort independently, coordinating requirements between mechanical, electrical, and security stakeholders to ensure system compatibility
- Delivered a production-ready mechanical solution now planned for integration into more than twenty fielded systems globally, supporting mission-critical hardware reliability

Executive Board Member

Jan 2026 – Present

Cornell Maker Club | Ithaca, NY

- Overseeing operations of a student-run makerspace laboratory providing access to 3D printers, laser cutters, electronics benches, machine tools, and fabrication equipment
- Training new members in safe operation of lab machinery and enforcing safety protocols to maintain a reliable and secure workspace
- Organizing and staffing engineering-focused events including hackathons, makeathons, and build nights, supporting rapid prototyping and teamwork under time constraints
- Conducting hands-on personal engineering projects within the lab, including robotics, mechatronics systems, and UAV components, gaining experience across mechanical, electrical, and fabrication workflows
- Coordinating with fellow board members to maintain equipment functionality, develop training materials, and support student engineering initiatives across campus

ADDITIONAL EXPERIENCE

Team Member

Jun 2024 – Aug 2024

Smoothie King | Denver, CO

- Delivered customer service in a fast-paced food service environment while preparing orders, handling transactions, and maintaining cleanliness
- Managed multiple responsibilities during busy shifts, demonstrating teamwork, efficiency, and attention to detail

AI and VR/XR Research Intern

May 2023 – Jul 2023

Access Mode | Denver, CO

- Supported communication, outreach, and planning for a tech accelerator through email, web correspondence, and social media engagement
- Researched AI and VR/XR industry applications through workshops, speaker sessions, and discussions with technology professionals
- Presented insights on AI, VR, and XR applications to local government officials, community members, and investors

Photography and Media Intern

Nov 2022 – May 2023

Black Family Advisory Council (BFAC) | Denver, CO

- Photographed community events and created promotional video content for social media outreach
- Supported digital engagement efforts by producing visual content for a district-affiliated advocacy organization

SKILLS

CAD & Simulation: SolidWorks (CAD, assemblies, drawings), SolidWorks Simulation (FEA), SolidWorks Flow (CFD), Autodesk Fusion

Fabrication & Machining: CNC mill & lathe (Cornell Red Apron certified), sheet metal fabrication, manual machining, soldering, composite surface work, 3D printing, laser cutting

Electromechanical Systems: BLDC, brushed DC, servo, and stepper motor integration; ESCs; custom gearboxes; sensors; harness fabrication

Microcontrollers & Electronics: Arduino, Raspberry Pi, STM32, Teensy, ESP32; circuit prototyping; multimeter & oscilloscope use

Programming & Controls: Python; Bluetooth data integration

Manufacturing & Lab Operations: Company PDM, tolerancing & dimensioning, material selection, assembly documentation, fabrication

Certification: FAA Part 107 Remote Pilot License

Languages: English (Fluent), Spanish (Conversational)